

Quiz 6

Instructions: There are totally 2 problems. You must show all of your work to receive a credit for a correct answer. No calculator is allowed in the quiz.

Problem 1. (5 points) Find all first partial derivatives of the function

$$f(x, y, z) = x - \sqrt{y^2 + z^2}.$$

$$f_x = \frac{\partial f}{\partial x} = 1$$

$$f_y = \frac{\partial f}{\partial y} = -\frac{1}{2}(y^2 + z^2)^{-\frac{1}{2}}(2y) = -\frac{y}{\sqrt{y^2 + z^2}}$$

$$f_z = \frac{\partial f}{\partial z} = -\frac{1}{2}(y^2 + z^2)^{-\frac{1}{2}}(2z) = -\frac{z}{\sqrt{y^2 + z^2}}$$

Problem 2. (5 points) Find all first and second partial derivatives of the function

$$g(x, y) = 3x^4y^3 + e^{2xy}.$$

$$g_x = 3 \cdot (4x^3)y^3 + e^{2xy} \cdot (2y) = 12x^3y^3 + 2ye^{2xy}$$

$$g_y = 3x^4(3y^2) + e^{2xy}(2x) = 9x^4y^2 + 2xe^{2xy}$$

$$g_{xx} = 36x^2y^3 + 4y^2e^{2xy}$$

$$g_{yy} = 18x^4y + 4x^2e^{2xy}$$

$$\begin{aligned} g_{xy} &= g_{yx} = 36x^3y^2 + 2e^{2xy} + 2y \cdot (2xe^{2xy}) \\ &= 36x^3y^2 + (2 + 4xy)e^{2xy} \end{aligned}$$